Matrix Manufacturing & Retail – Data Analytics Report

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Introduction

Businesses use data analytics to understand consumers and their behaviour, improve their performance and service (University of Bath, 2021). In our case, as a bicycle products business, at Matrix Manufacturing and Retail (MMR) can use business analytics tools and big data to more effectively support business decisions regarding product range and sales as well as our customer range and support. Data is an asset to any company. Currently, we have a lot of data at MMR but it is often fragmented, difficult to find and our reporting mechanisms are not fit for purpose. A data strategy can help us with that since it sets out the data governance principles (including documentation and quality standards as well as who is responsible for delivering such data) before any data analytics can be done.

Strategic planning is required as without the right internal and external data we cannot produce the qualitative and quantitative insights and compare them to MMR's key performance indicators (KPIs) such as profit margins or customer satisfaction rates. This involves project management of the different people involved in the organisation which have access to the internal data we require. We shall formally request only the relevant data to prevent overworking people and retrieving irrelevant information.

The data strategy for our MMR sets out how we intend to "collect, manage, govern, derive and utilize value from our data" (Howard and Kara, 2024) to address business goals. Our business goals are to improve product performance (frames and mountain sales) and customer loyalty.

Data strategy

The essential stakeholders are the data scientist, responsible for data collection and preparation, and the data analyst, responsible for producing actionable insights. In this case, my role overlaps with both. Then, there is my manager Srebrenka, responsible for delivering the presentation of the business analytics report to the senior executives next week.

Among our objectives are to explore our current product performance (general and bikes only) in terms of stock, current and prospective sales to suggest actionable changes to increase demand and revenue. Additionally, we want to explore our customer loyalty base to improve our customer experience and reduce our customer churn (retention).

A data pipeline needs to be established to make sure the data strategy can be carried out and there is sufficient support in all the components of this pipeline. This data pipeline will take in raw data (mostly internal in this case), transform it (clean it so it's usable) and deliver it to a data warehouse (a storage place) before it can be analysed (Stryker, 2026). The plan is to load and transform the data through SQL queries to be later analysed in Python. No additional new software to the existing ones (SQL, and Python) is required to deliver this report.

At the same time, we need to enforce ethical standards of data collection, analysis and storage. Complying with GDPR and other legal requirements of data protection and security is essential (Dessaints, 2025). Good data practices include informed consent to our service users (in this case learners and their legal guardians, if underage) about the types of data. All our customers when registering to our loyalty programme agreed to our terms and conditions, complying with GDPR and outlining how we could use their data.

The data I need for this report was all internal and it included the following SQL files requiring only the necessary columns to create the report:

- Customer data
  - Customer ID, first name, surname, location of purchase, number of purchases, loyalty status, date they joined the loyalty program (where applicable).
- Employee data
  - Employee ID, first name, surname, working pattern, location
- Products data
  - Product ID, product name, type, inventory, costs, number of sales, retail price

Business analytics capture our measures of success and align with align with the business goals stated previously regarding increasing revenue and reduce customer churn.

As Chen (2025) point out, there are four main types of business analytics available: descriptive, diagnostic, predictive and prescriptive (University of Bath, 2021). Each of these focus on different questions relevant to us. In the case of descriptive analytics, such as line, bar and pie charts, they tell us about the past of our business and often involve historical data (e.g. aggregated sales data). Among the uses of this type of analytics are to "analyse existing data to report on operational outcomes" (Dutta, 2024). One of the use cases of this type of analytics is to summarise customer loyalty by location (Fig 4).

Predictive data analytics, on the other hand, look at the future and try to determine the outcome (Maisel and Cokins, 2013). These type of analytics relies on machine learning and statistical modelling to identify opportunities but also risks (Cote, 2021b). Now with the rise of AI, these modelling techniques have become more advanced (unsupervised learning, large neural networks, etc.) and faster, as noted by IBM (2025). For the purpose of this report, this type of analytics provides us with insights into future market reach (which new type of customers can we attract in the upcoming financial year) based on our current customer base and their behaviour (how likely are they to purchase again with us).

The purpose of diagnostic analytics is to understand business outcomes, sometimes expected but more often unexpected or a problem of sorts. For example, in our case, we've noticed a decrease in the number of e-learners from lower-income families in the last quarter. Diagnostic analytics could tell us the reasons for such outcome by using statistical analysis tools like regression analysis, time series and related tests (Cote, 2021a). In our case, diagnostic analytics would help us understand how well are we doing in terms of the number of sales per product (Fig 1).

Lastly, prescriptive analytics focus on what the company could do in the future to keep improving. It looks at the future while providing actionable suggestions of how to deliver such outcomes (Cote, 2021c). In our case, these type of analytics could help us suggest which locations we should be increasing our marketing efforts. For example, locations where we have a high number Gold loyalty customers but not as good product sales, we could create targeted promotions (Fig 4). These are just some examples of the many benefits of both, having a data strategy and using business analytics.

Data creation and extraction

Using MySQL Workbench I collected and extracted the above data. Note that the original salesdatabase file did not include some of the columns of interest so I created them and assigned values accordingly. Then, I exported the corresponding tables (loyalty, general sales and bike sales) as csv files. See Additional notes section at the end of document for more details about this process.

Data analysis in Python

I imported the data into Python using the pandas module to analyse the data and create visualisations using seaborn and matplotlib modules. See Additional notes section at the end of document for more details about this process.

Product Sales

The first thing we notice is that few of our products have had a consistent sales in 2022 instead they have been fluctuating seasonally (see Figure 1). For example in between May and July, we had a spike of sales in accessories, clothing, frames, brakes and road bikes. Then, towards the end of the summer, we see that nuts and touring bikes increased in sales as well as our other product categories. Wheels and Tires were more frequently bought before the winter months.

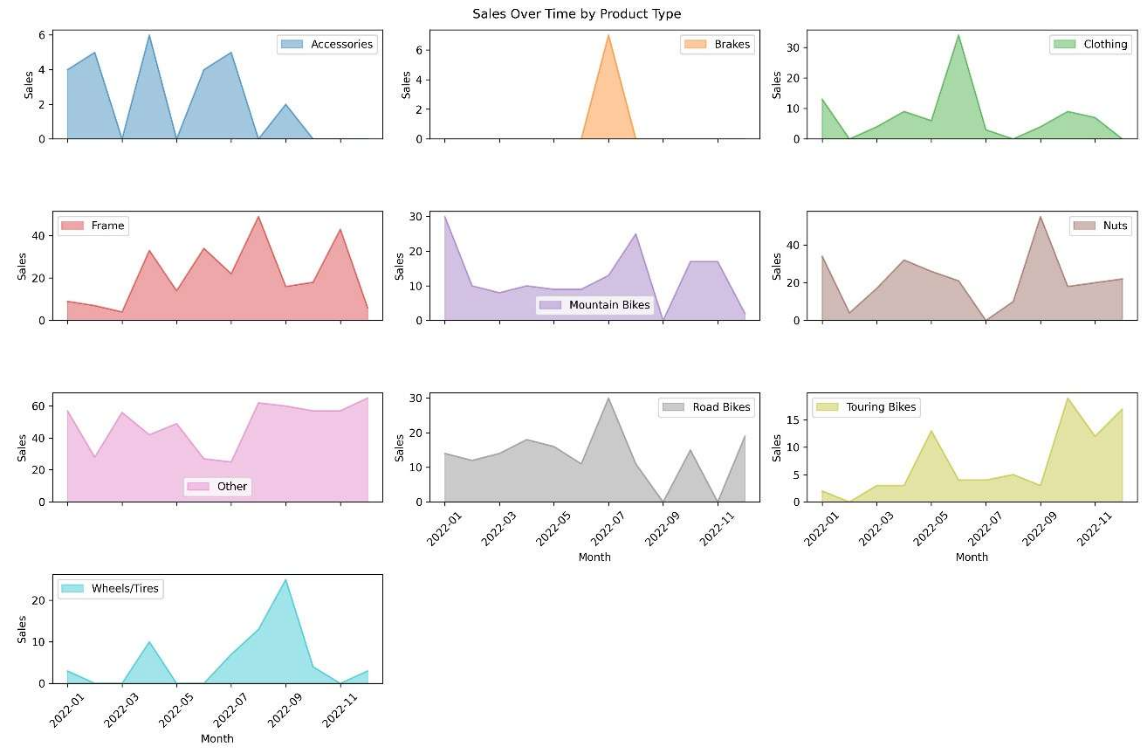

Figure 1. Shows the sales in 2022 for each the 10 product types in our shop.

Bike sales

Looking only at bike sales (see Figure 2), we find that mountain bikes were very popular at the beginning of the year but had a sharp decrease in sales over time with a slightly smaller peak in October. Touring bikes were among the least frequently sold bikes out of all our options but they became more popular towards the end of the year with a sharp increase in sales in October. Road bikes sales were seasonal with peaks at the start of Spring, Summer and Autumn suggesting people purchased them based on the weather.

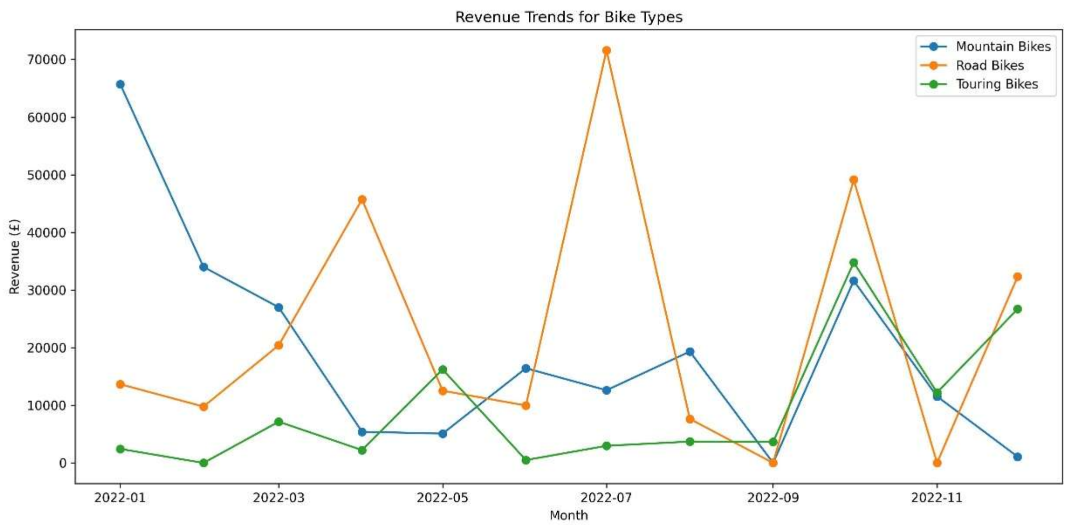

Figure 2. Annual sales of 2022 for each of the three bike types. Blue lines are mountain bikes, orange likes are road bikes and green lights are touring bikes.

Customer Loyalty

We introduced a loyalty programme and saw a growing number of customer joining the programme which is positive. This stayed constant across our five England and one Scottish locations, as shown in Figure 3. Manchester and London having the highest number of average purchases for non-registered customers compared to the rest of locations. Then both, Glasgow and Leeds had the largest average number of purchases by Gold customers although the average across locations was 45. London and Glasgow had slightly less purchases by Bronze customers and we find little differences across Silver customers in all our locations.

Figure 3. Average number of purchases per customer based on their location and loyalty status. Darker colours signal higher number of purchases.

To further understand our customer base and their purchasing activities, the below treemap takes into account the number of customers for each location (see Figure 4). We have a higher number of customers shopping in Newcastle and most of these are part of our loyalty program. On the other hand, areas like Birmingham and Glasgow have the fewest customers but the average purchase per customer of Gold tier customers is higher than all other locations (except Leeds which comes second before Birmingham). This suggests that while the market may be more limited we have access to reliable customer base.

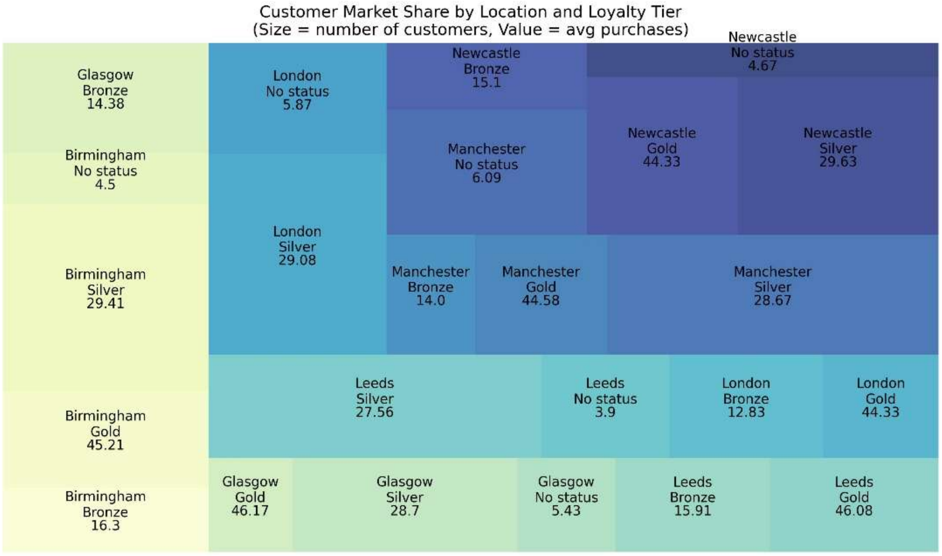

Figure 4. Average number of purchases by loyalty status and location accounting for the market share. Size represents number of customers per loyalty tier and location. Labels show average number of purchases.

Conclusion

Our product sales do not seem to have a clear trend which suggests we need to investigate further regarding our marketing and advertisements campaigns as we may be missing certain opportunities. Bike sales more specifically are doing well in the case of road bikes despite of the clear seasonal effect which suggests we should explore how to ensure both, mountain and touring bikes, grow in sales. Lastly in terms of our customer base and loyalty tiers. The programme is working well with healthy number of both, customers per tier and average number of purchases. Some locations like Manchester and Newcastle have a higher customer base but Birmingham and Glasgow have slightly better number of avg. purchases for Gold tier members which suggest a more loyal customer base there. We should explore how to take those insights into the other locations to continue growing.

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